

- a. Complete the entries in the table. Put the sums in the last row. What are the sample means $\bar{x}$ and $\bar{y}$?
- b. Calculate b_1 and b_2 using (2.7) and (2.8) and state their interpretation.
- c. Compute $\sum_{i=1}^5 x_i^2$, $\sum_{i=1}^5 x_i y_i$. Using these numerical values, show that $\sum (x_i - \bar{x})^2 = \sum x_i^2 - N\bar{x}^2$ and $\sum (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - N\bar{x}\bar{y}$.
- d. Use the least squares estimates from part (b) to compute the fitted values of y , and complete the remainder of the table below. Put the sums in the last row.
Calculate the sample variance of y , $s_y^2 = \sum_{i=1}^N (y_i - \bar{y})^2 / (N - 1)$, the sample variance of x , $s_x^2 = \sum_{i=1}^N (x_i - \bar{x})^2 / (N - 1)$, the sample covariance between x and y , $s_{xy} = \sum_{i=1}^N (y_i - \bar{y})(x_i - \bar{x}) / (N - 1)$, the sample correlation between x and y , $r_{xy} = s_{xy} / (s_x s_y)$ and the coefficient of variation of x , $CV_x = 100(s_x / \bar{x})$. What is the median, 50th percentile, of x ?

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$
3	4				
2	2				
1	3				
-1	1				
0	0				
$\sum x_i =$	$\sum y_i =$	$\sum \hat{y}_i =$	$\sum \hat{e}_i =$	$\sum \hat{e}_i^2 =$	$\sum x_i \hat{e}_i =$

- e. On graph paper, plot the data points and sketch the fitted regression line $\hat{y}_i = b_1 + b_2 x_i$.
- f. On the sketch in part (e), locate the point of the means $(\bar{x}, \bar{y})$. Does your fitted line pass through that point? If not, go back to the drawing board, literally.
- g. Show that for these numerical values $\bar{y} = b_1 + b_2 \bar{x}$.
- h. Show that for these numerical values $\bar{\hat{y}} = \bar{y}$, where $\bar{\hat{y}} = \sum \hat{y}_i / N$.
- i. Compute $\hat{\sigma}^2$.
- j. Compute $\text{var}(b_2 | \mathbf{x})$ and $\text{se}(b_2)$.

$$a. \sum x_i = 3 + 2 + 1 + (-1) + 0 = 5$$

$$\sum y_i = 4 + 2 + 3 + 1 + 0 = 10$$

$$\bar{x} = \frac{5}{5} = 1 \quad \bar{y} = \frac{10}{5} = 2$$

$$c. x_i^2: 9, 4, 1, 1, 0 \Rightarrow \sum_{i=1}^5 x_i^2 = 15$$

$$x_i y_i: 12, 4, 3, -1, 0 \Rightarrow \sum_{i=1}^5 x_i y_i = 18$$

$$\sum (x_i - \bar{x})^2 = \sum x_i^2 - N\bar{x}^2$$

$$10 = 15 - 5(1^2) = 10$$

左右相等, 得證

$$\sum (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - N\bar{x}\bar{y}$$

$$8 = 18 - 5(1)(2) = 8$$

左右相等, 得證 *

$$b. \quad b_2 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{8}{10} = 0.8$$

$$b_1 = \bar{y} - b_2 \bar{x} = 2 - 0.8(1) = 1.2$$

b_2 (slope) = 0.8 when x increase 1 unit, the expected

average value of y increase 0.8 unit.

b_1 (intercept) = 1.2 when $x=0$, the expected average value of y is 1.2.

d. $\hat{y} = 1.2 + 0.8x$

$\hat{e} = y - \hat{y}$

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$
3	4	$1.2 + 0.8(3) = 3.6$	$4 - 3.6 = 0.4$	0.16	$3(0.4) = 1.2$
2	2	$1.2 + 1.6 = 2.8$	$2 - 2.8 = -0.8$	0.64	$2(-0.8) = -1.6$
1	3	$1.2 + 0.8 = 2$	$3 - 2 = 1$	1	$1(1) = 1$
-1	1	$1.2 - 0.8 = 0.4$	$1 - 0.4 = 0.6$	0.36	$-1(0.6) = -0.6$
0	0	$1.2 + 0 = 1.2$	$0 - 1.2 = -1.2$	1.44	$0(-1.2) = 0$
$\sum x_i = 5$	$\sum y_i = 10$	$\sum \hat{y}_i = 10$	$\sum \hat{e}_i = 0$	$\sum \hat{e}_i^2 = 3.6$	$\sum x_i \hat{e}_i = 0$

$$\sum (y_i - \bar{y})^2 = 2^2 + 0 + 1^2 + (-1)^2 + (-2)^2 = 10$$

$$s_y^2 = \frac{\sum (y_i - \bar{y})^2}{n-1} = \frac{10}{4} = 2.5$$

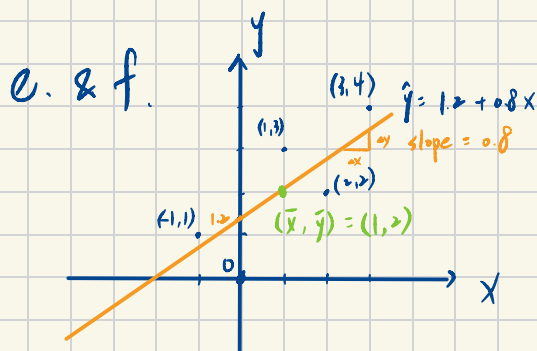
$$s_x^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} = \frac{10}{4} = 2.5$$

$$s_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1} = \frac{8}{4} = 2$$

$$r_{xy} = \frac{s_{xy}}{s_x s_y} = \frac{2}{\sqrt{2.5} \sqrt{2.5}} = \frac{2}{2.5} = 0.8$$

$$CV_x = 100 \times \left(\frac{s_x}{\bar{x}} \right) = 100 \times \left(\frac{\sqrt{2.5}}{1} \right) \approx 100 \times 1.5811 = 158.11$$

Median of $x \Rightarrow (-1, 0, (1), 2, 3) \therefore$ median of x is 1.



$$\therefore b_1 = \bar{y} - b_0 \bar{x}$$

$\therefore$ the regression line indeed pass through the point $(\bar{x}, \bar{y}) = (1, 2)$ #

$$g. \quad \bar{y} = b_1 + b_2 \bar{x}$$

$$2 = 1.2 + 0.8(1) \quad \text{左右相等, 得證} *$$

$$h. \quad \sum \hat{y}_i = 10$$

$$\bar{\hat{y}} = \frac{\sum \hat{y}_i}{N} = \frac{10}{5} = 2 = \bar{y}$$

$$\therefore \bar{y} = \bar{\hat{y}} *$$

$$i. \quad \hat{\sigma}^2 = \frac{\sum \hat{e}_i^2}{N-2} = \frac{3.6}{5-2} = \frac{3.6}{3} = 1.2 *$$

$$j. \quad \widehat{\text{var}}(b_2|x) = \frac{\hat{\sigma}^2}{\sum (x_i - \bar{x})^2} = \frac{1.2}{10} = 0.12 *$$

$$\text{se}(b_2) = \sqrt{0.12} \approx 0.3464 *$$

hw0303

2026-03-04

```
library(readxl)
```

```
## Warning: 套件 'readxl' 是用 R 版本 4.2.3 來建造的
```

```
star5_small <- read_excel("C:/finance ecomatrics/data/star5_small.xlsx")
head(star5_small)
```

```
## # A tibble: 6 × 7
##   aide mathscore readscore regular small tchexper totalscore
##   <dbl>    <dbl>    <dbl>    <dbl> <dbl>    <dbl>    <dbl>
## 1     0      528      461         0     1      13      989
## 2     0      454      436         1     0       8      890
## 3     1      463      395         0     0       8      858
## 4     1      449      430         0     0       3      879
## 5     0      478      421         0     1      15      899
## 6     1      536      445         0     0      11      981
```

```
summary(star5_small)
```

```
##      aide      mathscore      readscore      regular
##  Min.   :0.0000  Min.   :339.0  Min.   :358.0  Min.   :0.0000
## 1st Qu.:0.0000 1st Qu.:454.0 1st Qu.:414.0 1st Qu.:0.0000
##  Median :0.0000 Median :478.0 Median :433.0 Median :0.0000
##  Mean    :0.3542 Mean    :485.9 Mean    :435.8 Mean    :0.3433
## 3rd Qu.:1.0000 3rd Qu.:513.0 3rd Qu.:451.0 3rd Qu.:1.0000
##  Max.    :1.0000 Max.    :626.0 Max.    :627.0 Max.    :1.0000
##
##      small      tchexper      totalscore
##  Min.   :0.0000  Min.   : 0.000  Min.   : 727.0
## 1st Qu.:0.0000 1st Qu.: 5.000 1st Qu.: 870.0
##  Median :0.0000 Median : 9.000 Median : 915.0
##  Mean    :0.3025 Mean    : 9.126 Mean    : 921.6
## 3rd Qu.:1.0000 3rd Qu.:13.000 3rd Qu.: 963.0
##  Max.    :1.0000 Max.    :27.000 Max.    :1253.0
##
##      NA's :6
```

2.22 (a)

```
data_a <- subset(star5_small, small==1 | regular==1)
model_a <- lm(totalscore ~ small, data = data_a)
summary(model_a)
```

```
##
## Call:
## lm(formula = totalscore ~ small, data = data_a)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -189.44  -50.44   -9.44   43.06  324.38
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  916.442      3.675  249.396   <2e-16 ***
## small        12.175       5.369   2.268   0.0236 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 74.59 on 773 degrees of freedom
## Multiple R-squared:  0.006608,    Adjusted R-squared:  0.005323
## F-statistic: 5.142 on 1 and 773 DF,  p-value: 0.02363
```

Based on the R output provided, the estimated regression equation is:

$$TOTALSCORE = 916.442 + 12.175 \times SMALL$$

1. Intercept (β_1): 916.442 This represents the estimated average total score for students in regular-sized classes (where small = 0). 2. Slope for small (β_2): 12.175 On average, students assigned to small classes score 12.175 points higher than students in regular-sized classes

Conclusion: We conclude that class size has a statistically significant effect on student learning. Specifically, being in a small class positively impacts students' academic performance, leading to higher total scores on average.

2.22(b)

```
model_b_read <- lm(readscore ~ small, data = data_a)
summary(model_b_read)
```

```
##
## Call:
## lm(formula = readscore ~ small, data = data_a)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -74.665 -21.590  -2.665  16.410  194.335
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  432.665      1.488  290.760 < 2e-16 ***
## small        6.924       2.174   3.185  0.00151 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 30.2 on 773 degrees of freedom
## Multiple R-squared:  0.01295,    Adjusted R-squared:  0.01167
## F-statistic: 10.14 on 1 and 773 DF,  p-value: 0.001507
```

```
model_b_math <- lm(mathscore ~ small, data = data_a)
summary(model_b_math)
```

```
##
## Call:
## lm(formula = mathscore ~ small, data = data_a)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -144.777 -35.028  -5.028   29.223  142.223
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  483.777      2.449  197.545 <2e-16 ***
## small        5.251       3.578   1.467   0.143
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 49.71 on 773 degrees of freedom
## Multiple R-squared:  0.002778,    Adjusted R-squared:  0.001488
## F-statistic: 2.153 on 1 and 773 DF,  p-value: 0.1427
```

for the Reading Score Model, estimated equation:

$$READSCORE = 432.665 + 6.924 \times SMALL$$

1. Intercept (432.665): The estimated average reading score for students in regular-sized classes is 432.665.
2. Slope for small (6.924): Students in small classes score, on average, 6.924 points higher in reading than those in regular classes. The p-value of 0.00151 indicates statistical significance.

for the Math Score Model, estimated equation:

$$MATHSCORE = 483.777 + 5.251 \times SMALL$$

1. Intercept (483.777): The estimated average math score for students in regular-sized classes is 483.777.

2. Slope for small (5.251): Students in small classes score, on average, 5.251 points higher in math than those in regular classes. The p-value of 0.143 indicates no statistical significance.

Conclusion: This suggests that class size reduction may be effective in improving reading skills, but fails to provide significant improvements in quantitative and math skills.

2.22(c)

```
data_c <- subset(star5_small, aide==1 | regular==1)
model_c <- lm(totalscore ~ aide, data = data_c)
summary(model_c)
```

```
##
## Call:
## lm(formula = totalscore ~ aide, data = data_c)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -191.748  -50.442   -5.442    40.558   312.558
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  916.442      3.559  257.529  <2e-16 ***
## aide         4.306       4.994   0.862   0.389
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 72.23 on 835 degrees of freedom
## Multiple R-squared:  0.0008898, Adjusted R-squared:  -0.0003068
## F-statistic: 0.7436 on 1 and 835 DF, p-value: 0.3888
```

Based on the R output provided, the estimated regression equation is:

$$TOTALSCORE = 916.442 + 4.306 \times AIDE$$

1. Intercept (γ_1): 916.442 This represents the estimated average total score for students in regular-sized classes (where aide = 0). 2. Slope for small (γ_2): 4.306 students in regular-sized classes with a teacher aide score, on average, a mere 4.306 points higher than those in regular classes without an aide. But the p-value of 0.389 indicates no statistical significance.

Conclusion: Because the p-value is not significant, we fail to reject the null hypothesis. We conclude that adding a full-time teacher aide to a regular-sized classroom has no statistically significant effect on students' overall learning (total scores).

2.22(d)

```
model_d_read <- lm(readscore ~ aide, data = data_c)
summary(model_d_read)
```

```
##
## Call:
## lm(formula = readscore ~ aide, data = data_c)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -74.665 -21.536  -2.536  15.464 194.335
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  432.665      1.460  296.341  <2e-16 ***
## aide         2.871       2.049   1.401   0.161
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 29.64 on 835 degrees of freedom
## Multiple R-squared:  0.002347, Adjusted R-squared:  0.001152
## F-statistic: 1.964 on 1 and 835 DF, p-value: 0.1615
```

```
model_d_math <- lm(mathscore ~ aide, data = data_c)
summary(model_d_math)
```

```
##
## Call:
## lm(formula = mathscore ~ aide, data = data_c)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -146.212 -31.212  -5.777   29.223  142.223
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  483.777      2.355  205.464  <2e-16 ***
## aide         1.435       3.304   0.434   0.664
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 47.79 on 835 degrees of freedom
## Multiple R-squared:  0.0002258, Adjusted R-squared: -0.0009715
## F-statistic: 0.1886 on 1 and 835 DF, p-value: 0.6642
```

for the Reading Score Model, estimated equation:

$$READSCORE = 432.665 + 2.871 \times AIDE$$

1. Intercept (432.665): The estimated average reading score for students in regular-sized classes without an aide is 432.665.
2. Slope for aide (2.871): Having a teacher aide increases the reading score by an average of 2.871 points. However, with a p-value of 0.161, this effect is not statistically significant.

for the Math Score Model, estimated equation:

$$MATHSCORE = 483.777 + 1.435 \times AIDE$$

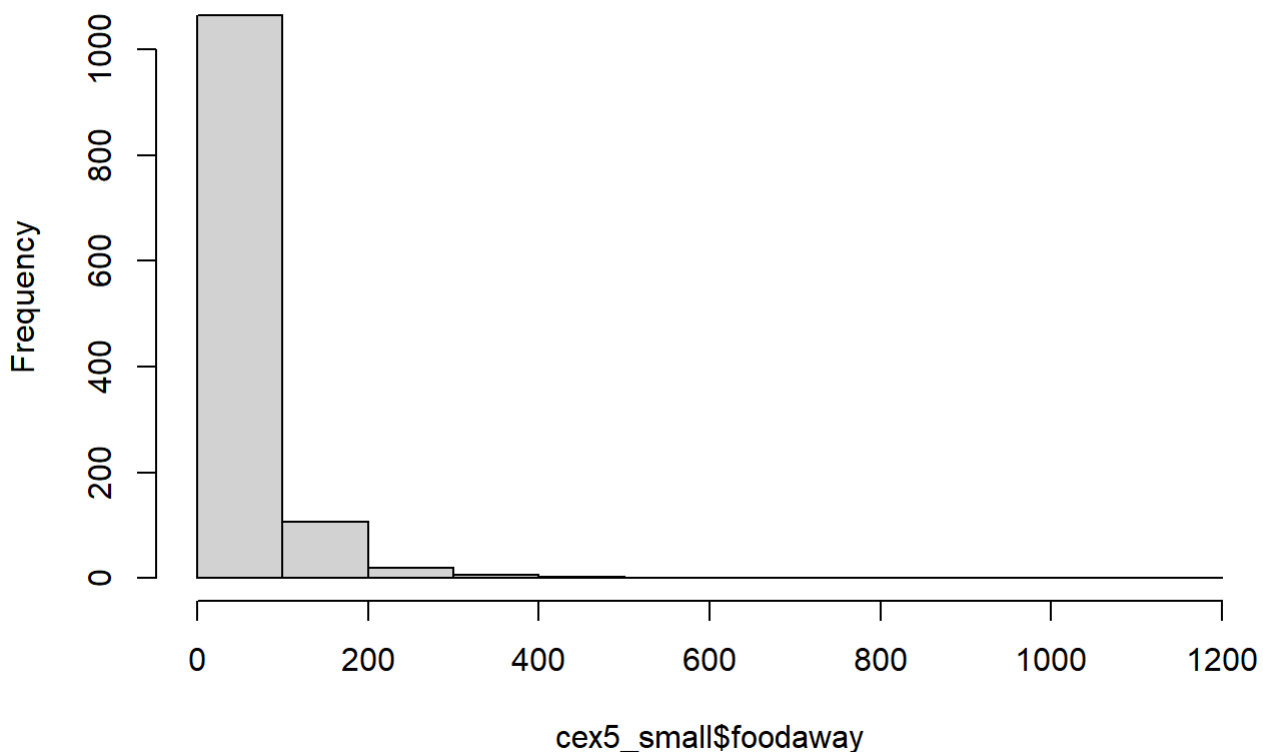
1. Intercept (483.777): The estimated average math score for students in regular-sized classes without an aide is 483.777.
2. Slope for aide (1.435): Having a teacher aide increases the math score by 1.435 points on average. With a p-value of 0.664, this effect is also not statistically significant.

Conclusion: Both p-values (0.161 and 0.664) are highly insignificant, failing to reject the null hypothesis. Therefore, we cannot conclude that adding a teacher aide has impact on either reading or math scores.

2.25(a)

```
library(readxl)
cex5_small <- read_excel("C:/finance ecomatrics/data/cex5_small.xlsx")
hist(cex5_small$foodaway)
```

Histogram of cex5_small\$foodaway



```
summary(cex5_small$foodaway)
```

##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
##	0.00	12.04	32.55	49.27	67.50	1179.00

mean =49.27 median =32.55 25th percentile =12.04 75th percentile =67.50

2.25(b)

```
d1 <- subset(cex5_small, advanced == 1)
mean(d1$foodaway)
```

```
## [1] 73.15494
```

```
median(d1$foodaway)
```

```
## [1] 48.15
```

```
d2 <- subset(cex5_small, college == 1)
mean(d2$foodaway)
```

```
## [1] 48.59718
```

```
median(d2$foodaway)
```

```
## [1] 36.11
```

```
d3 <- subset(cex5_small, advanced == 0 & college == 0)
mean(d3$foodaway)
```

```
## [1] 39.01017
```

```
median(d3$foodaway)
```

```
## [1] 26.02
```

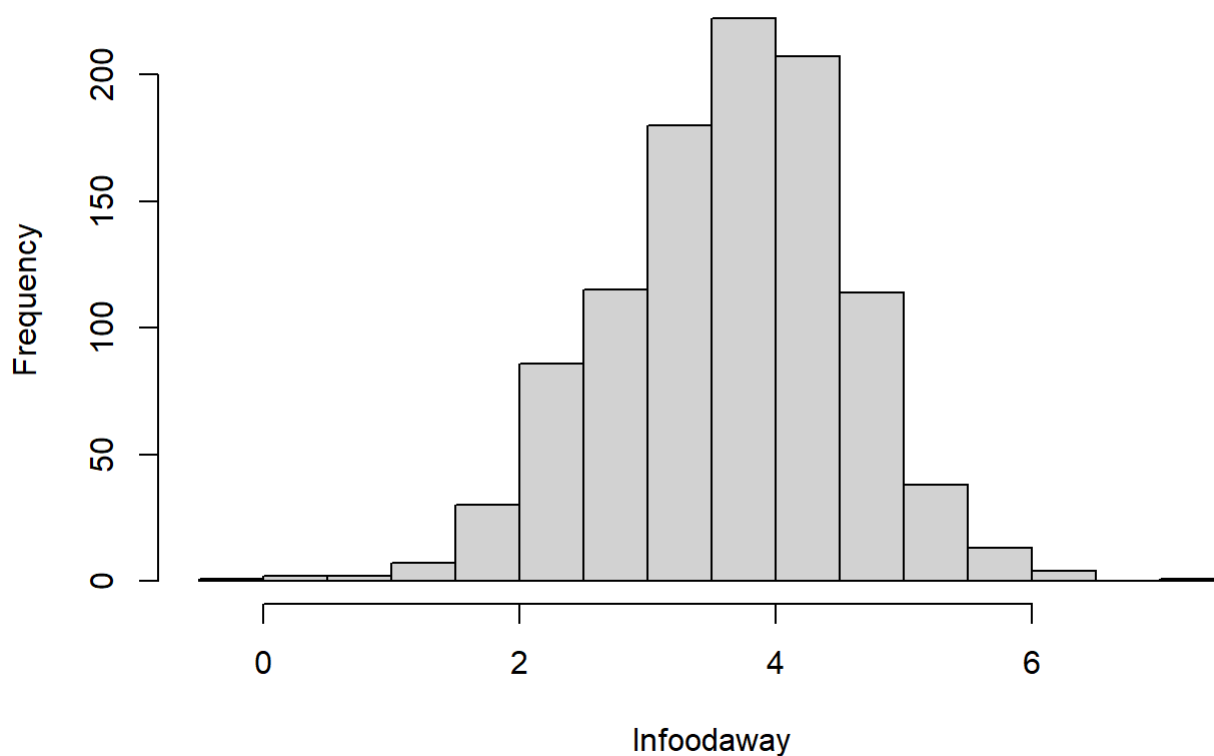
2.25(c)

```
lnfoodaway <- log(cex5_small$foodaway)
lnfoodaway[is.infinite(lnfoodaway)] <- NA
sum(is.na(lnfoodaway))
```

```
## [1] 178
```

```
hist(lnfoodaway)
```

Histogram of lnfoodaway



```
summary(lnfoodaway)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NA's
## -0.3011  3.0759  3.6865  3.6508  4.2797  7.0724     178
```

There are 178 fewer values of $\ln(\text{FOODAWAY})$. Observations with zero expenditure are dropped because the log transformation is undefined at zero. So $\ln(\text{foodaway})$ has less observations.

2.25(d)

```
cex5_small$lnfoodaway <- log(cex5_small$foodaway)
cex5_small$lnfoodaway[is.infinite(cex5_small$lnfoodaway)] <- NA
regmod <- lm(lnfoodaway ~ income, data = cex5_small, , na.action = na.exclude)
summary(regmod)
```

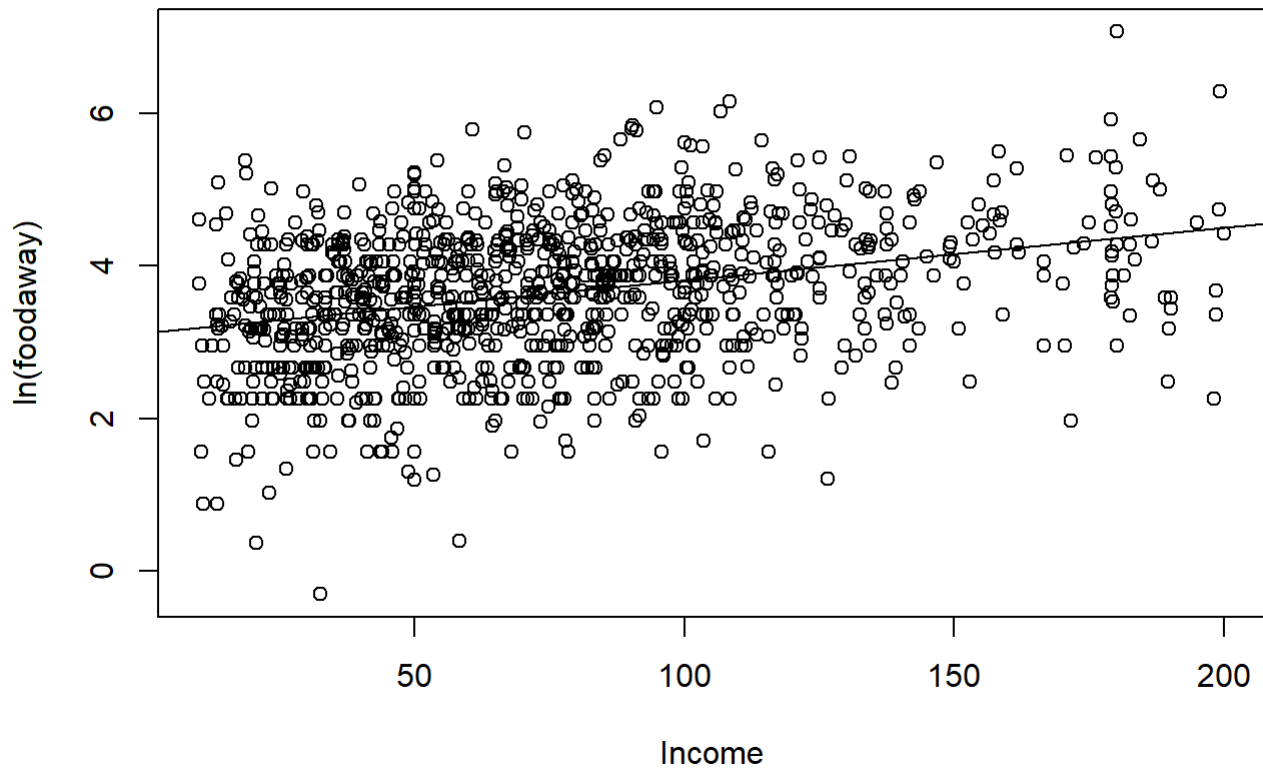
```
##
## Call:
## lm(formula = lnfoodaway ~ income, data = cex5_small, na.action = na.exclude)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.6547 -0.5777  0.0530  0.5937  2.7000
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.1293004   0.0565503   55.34  <2e-16 ***
## income       0.0069017   0.0006546   10.54  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8761 on 1020 degrees of freedom
##   (因為不存在，178 個觀察量被刪除了)
## Multiple R-squared:  0.09826,    Adjusted R-squared:  0.09738
## F-statistic: 111.1 on 1 and 1020 DF,  p-value: < 2.2e-16
```

each additional \$100 household income increases food away expenditures per person of about 0.69%, other factors held constant.

2.25(e)

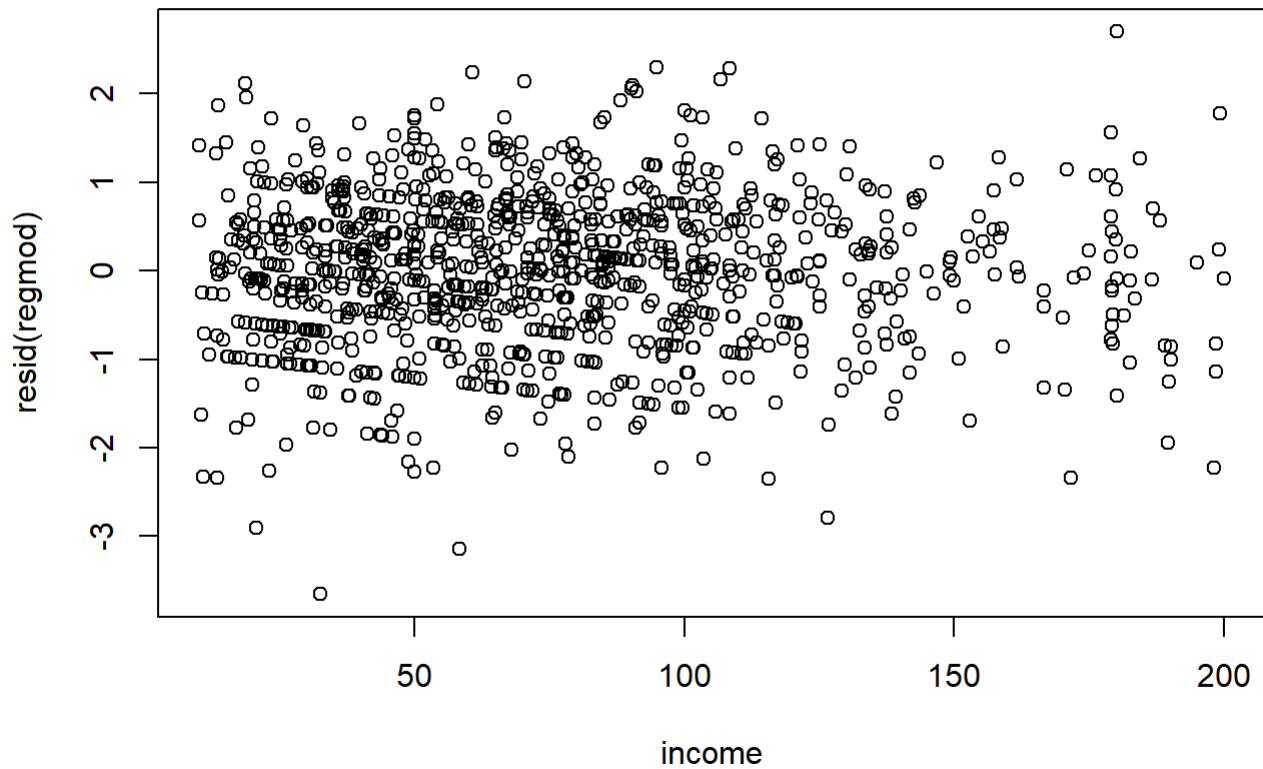
```
plot(cex5_small$income, cex5_small$lnfoodaway, xlab = "Income", ylab = "ln(foodaway)", main =
"Scatterplot + Fitted Line")
abline(regmod)
```

Scatterplot + Fitted Line



2.25(f)

```
plot(resid(regmod) ~income, data = cex5_small)
```



The OLS residuals do appear randomly distributed with no obvious patterns. There are fewer observations at higher incomes, so there is more white space.